

Thermal Response Functions for Chikungunya Transmission Risk

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Introduction

Before building the reproduction number model, I first examine the individual temperature-dependent functions used in the model. The purpose is to make sure each function behaves as expected over the temperature range. After checking and approving these functions separately, I will use them in the full $R_0(T)$ model. We use the Mordecai et al. model as the baseline. In this model, the basic reproduction number depends on temperature:

$$R_0(T) = \sqrt{\frac{a(T)^2 b(T) c(T) e^{-\left(\frac{\mu(T)}{PDR(T)}\right)} EFD(T) p_{EA}(T) MDR(T)}{Nr\mu(T)^3}}$$

The temperature-dependent functions are:

$$a(T) = \text{biting rate}$$

$$b(T) = \text{vector-to-human transmission probability}$$

$$c(T) = \text{human-to-vector transmission probability}$$

$$\mu(T) = \text{adult mosquito mortality rate}$$

$$PDR(T) = \text{parasite development rate}$$

$$EFD(T) = \text{eggs per female per day}$$

$$p_{EA}(T) = \text{egg-to-adult survival probability}$$

$$MDR(T) = \text{mosquito development rate}$$

Warning: package 'ggplot2' was built under R version 4.2.3

Biting rate

a is the per-mosquito biting rate.

- For the Scott et al. (2000) *Ae. aegypti* function, values outside 21–32 C are set to zero.

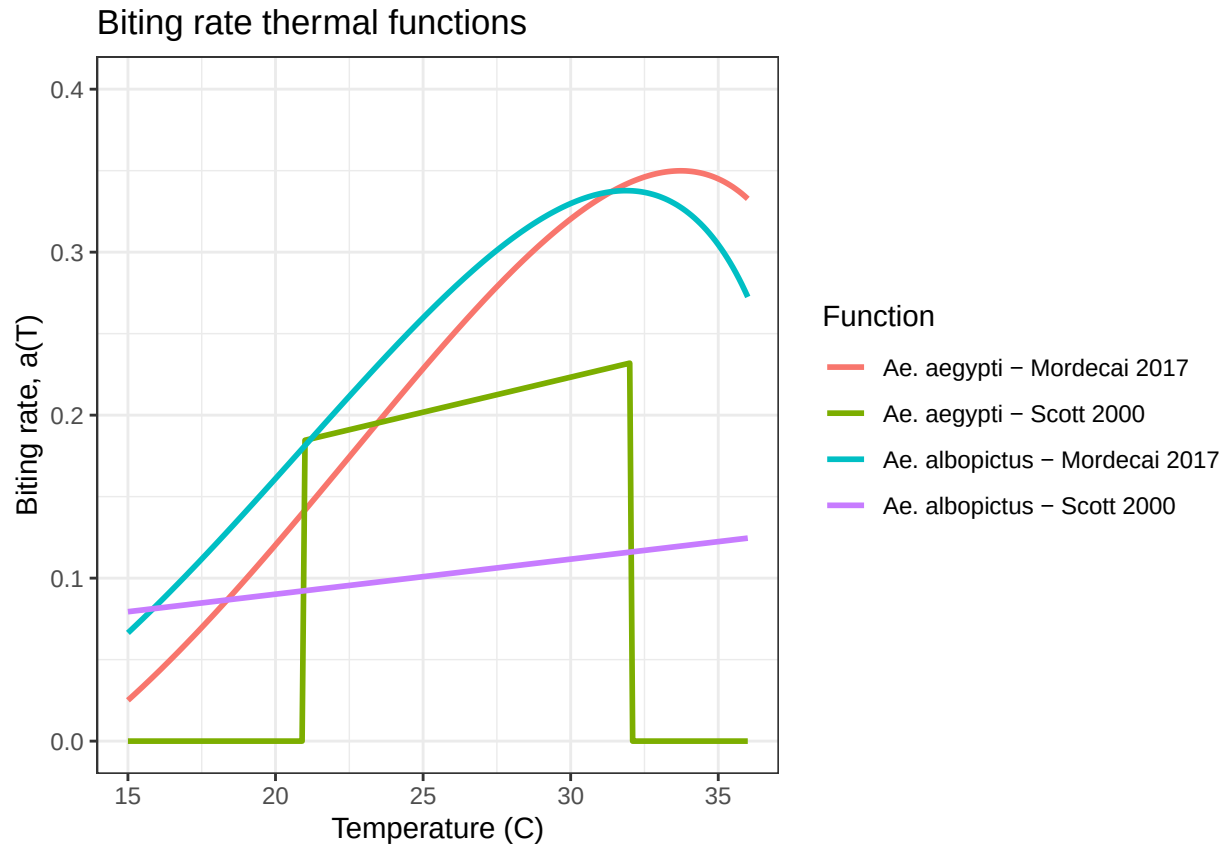


Figure 1: Biting rate thermal functions

Transmission Probability (Vector -> Human)

b is the proportion of infectious bites that infect susceptible humans.

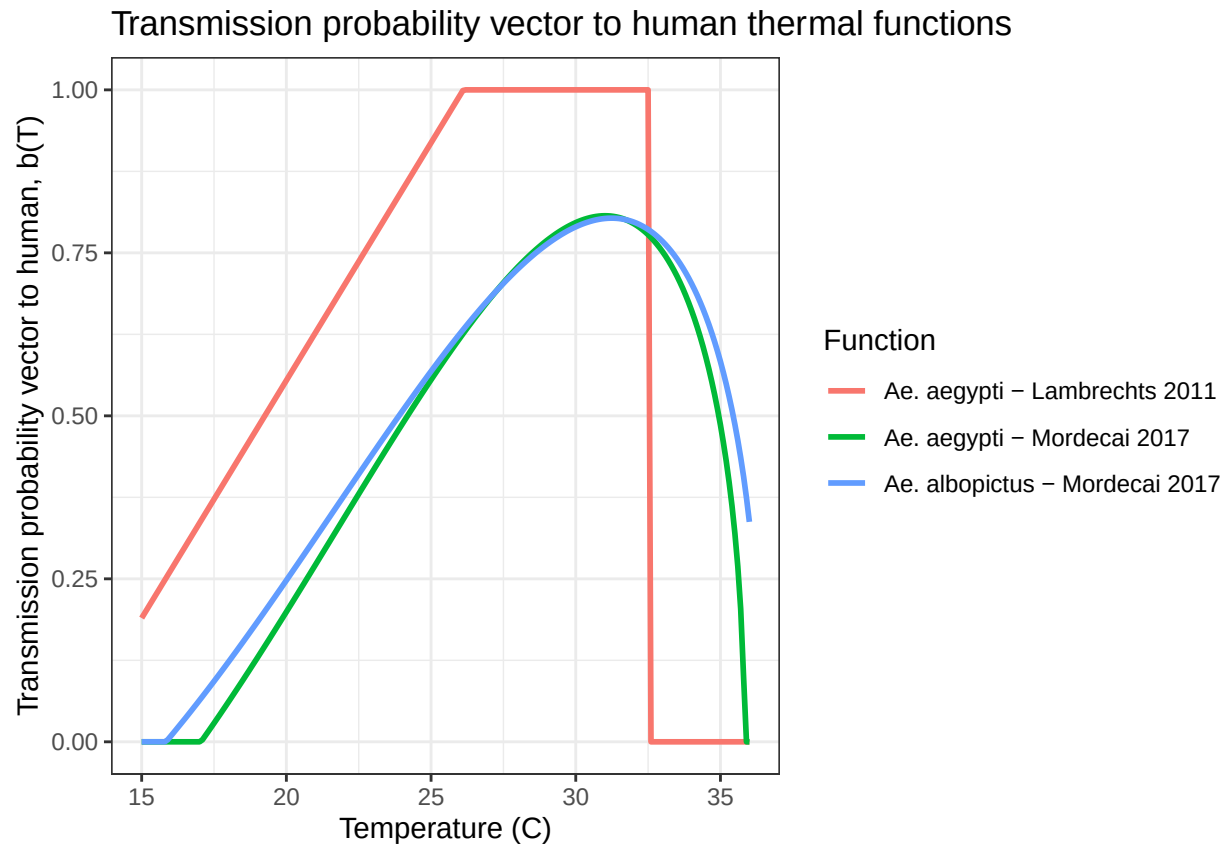


Figure 2: Transmission probability vector to human thermal functions

Transmission Probability (Human $\rightarrow$ Vector)

c is the proportion of bites on infected humans that infect previously uninfected mosquitoes. (i.e., b^*c = vector competence).

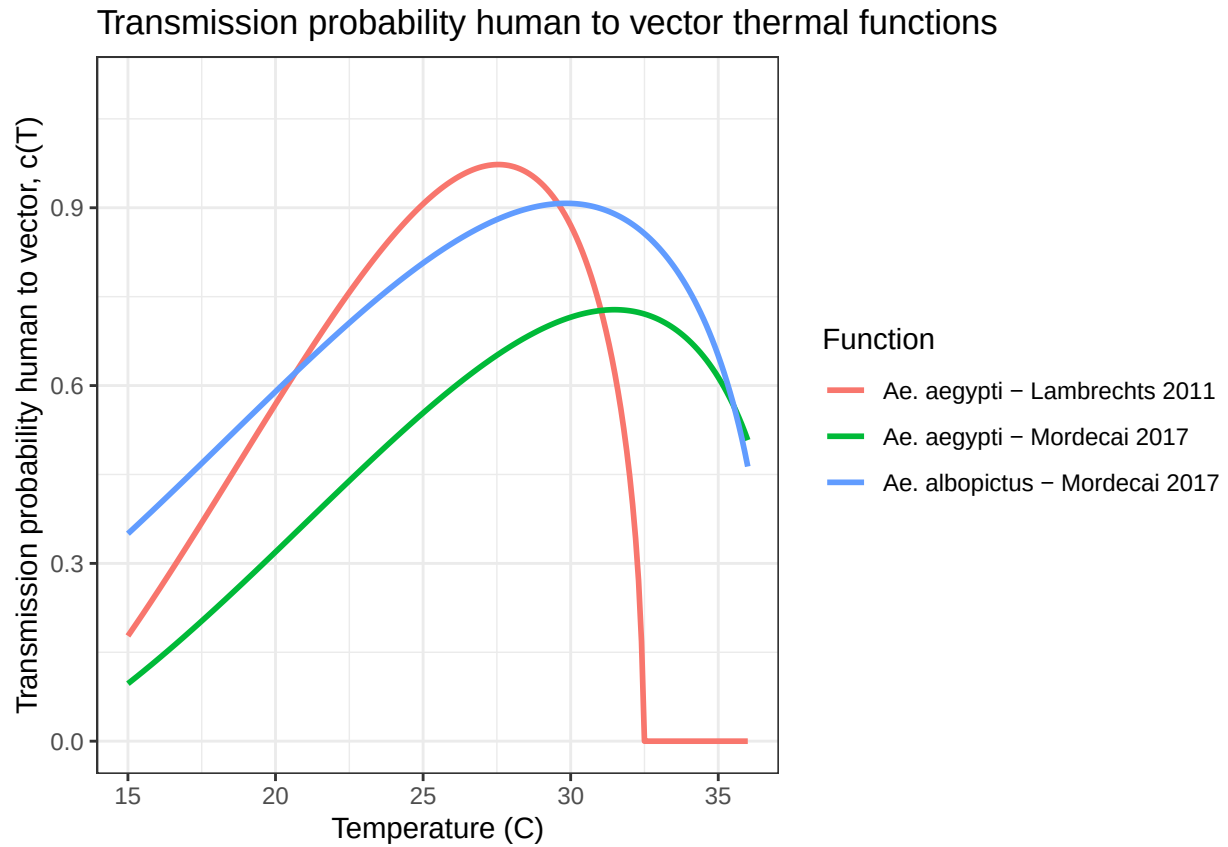


Figure 3: Transmission probability human to vector thermal functions

Adult Mosquito Lifespan

$1/\mu(T)$ is the adult mosquito lifespan.

- The correct *Ae. albopictus* lifespan formula is $-1.43(T - 13.41)(T - 31.51)$, not $-1.43T(T - 13.41)(T - 31.51)$.

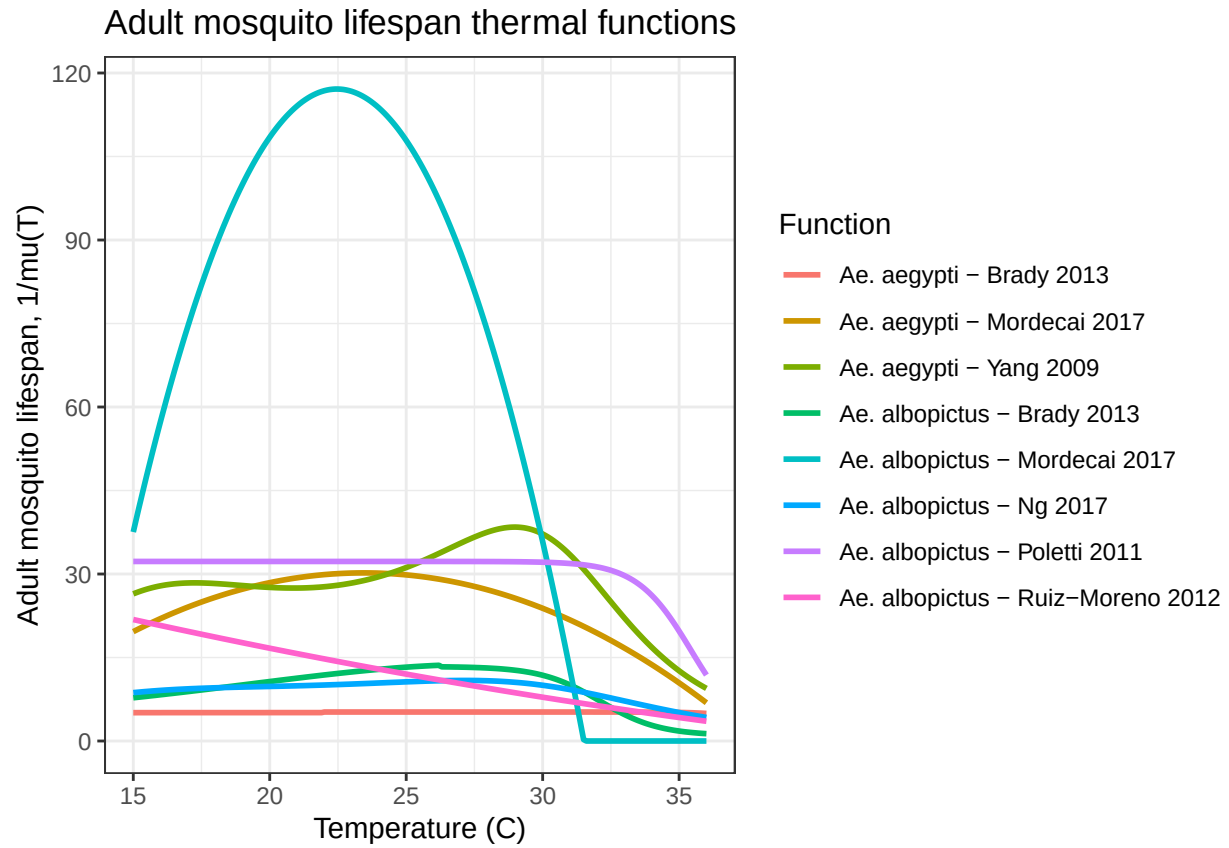


Figure 4: Adult mosquito lifespan thermal functions

Parasite Development Rate

PDR is the parasite development rate (i.e., the inverse of the extrinsic incubation period, the time required between a mosquito biting an infected host and becoming infectious)

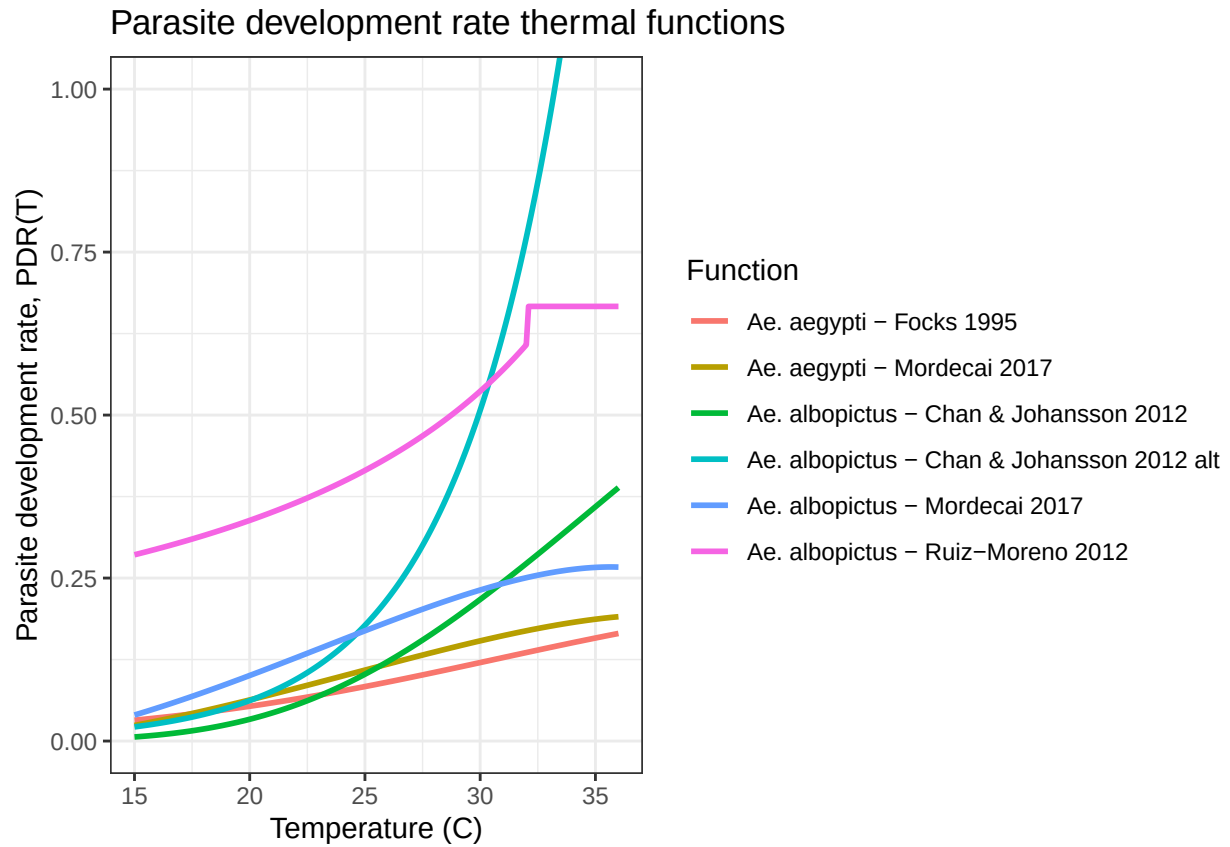


Figure 5: Parasite development rate thermal functions

Eggs per Female per Day (Fecundity)

EFD (*Ae. aegypti*) is the number of eggs produced per female mosquito per day and TFD (*Ae. albopictus*) is number of eggs laid per female per gonotrophic cycle (number/female)

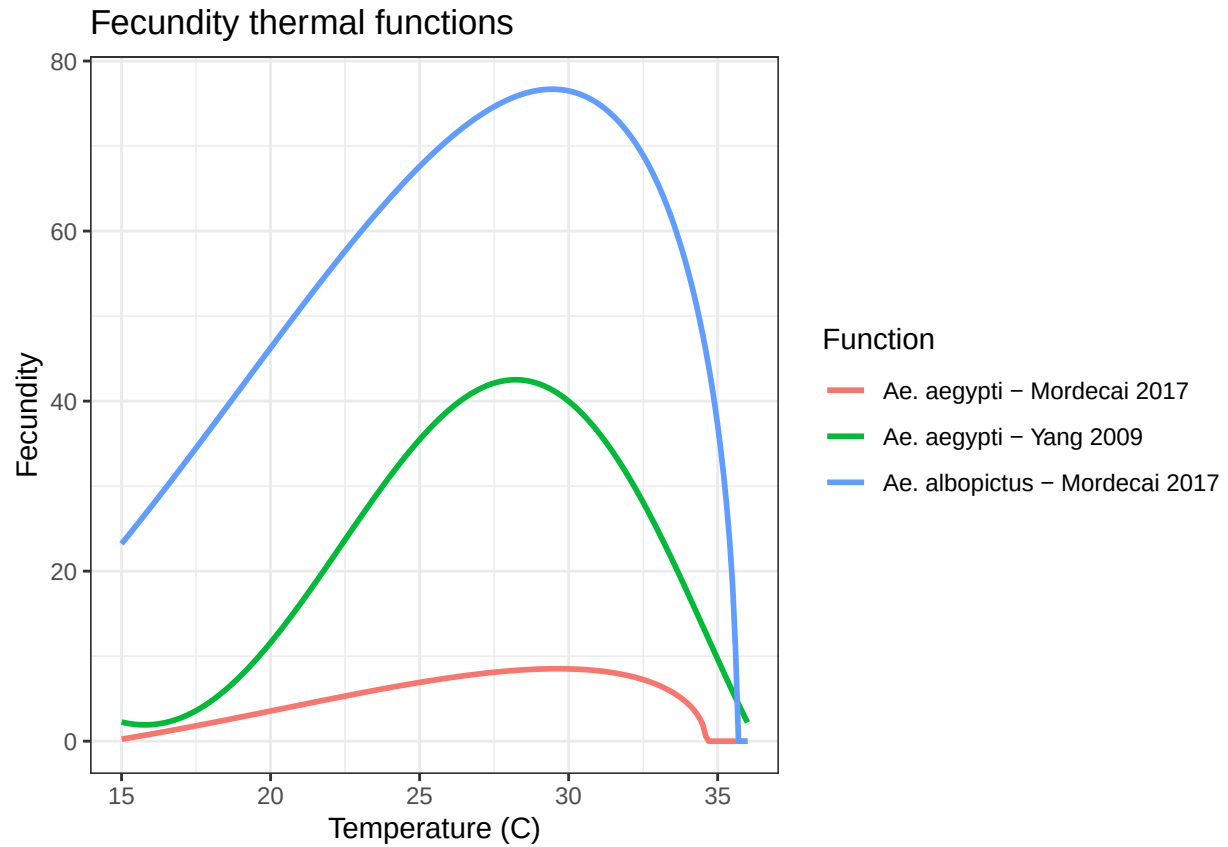


Figure 6: Fecundity thermal functions

Egg-Adult Survival Probability

p_{EA} is the mosquito egg-to-adult survival probability

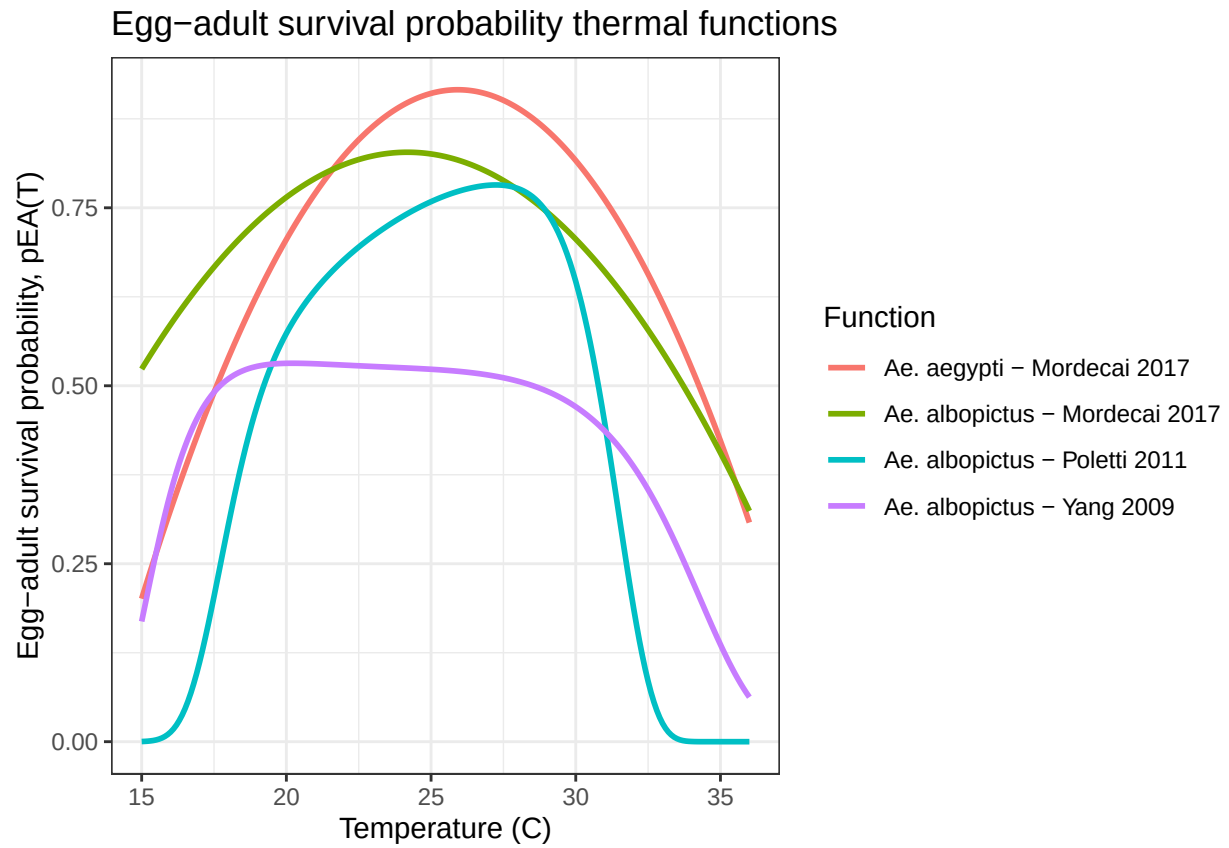


Figure 7: Egg-adult survival probability thermal functions

Mosquito Immature Development Rate (Aquatic Maturation)

MDR is the mosquito immature development rate (i.e., the inverse of the egg-to-adult development time)

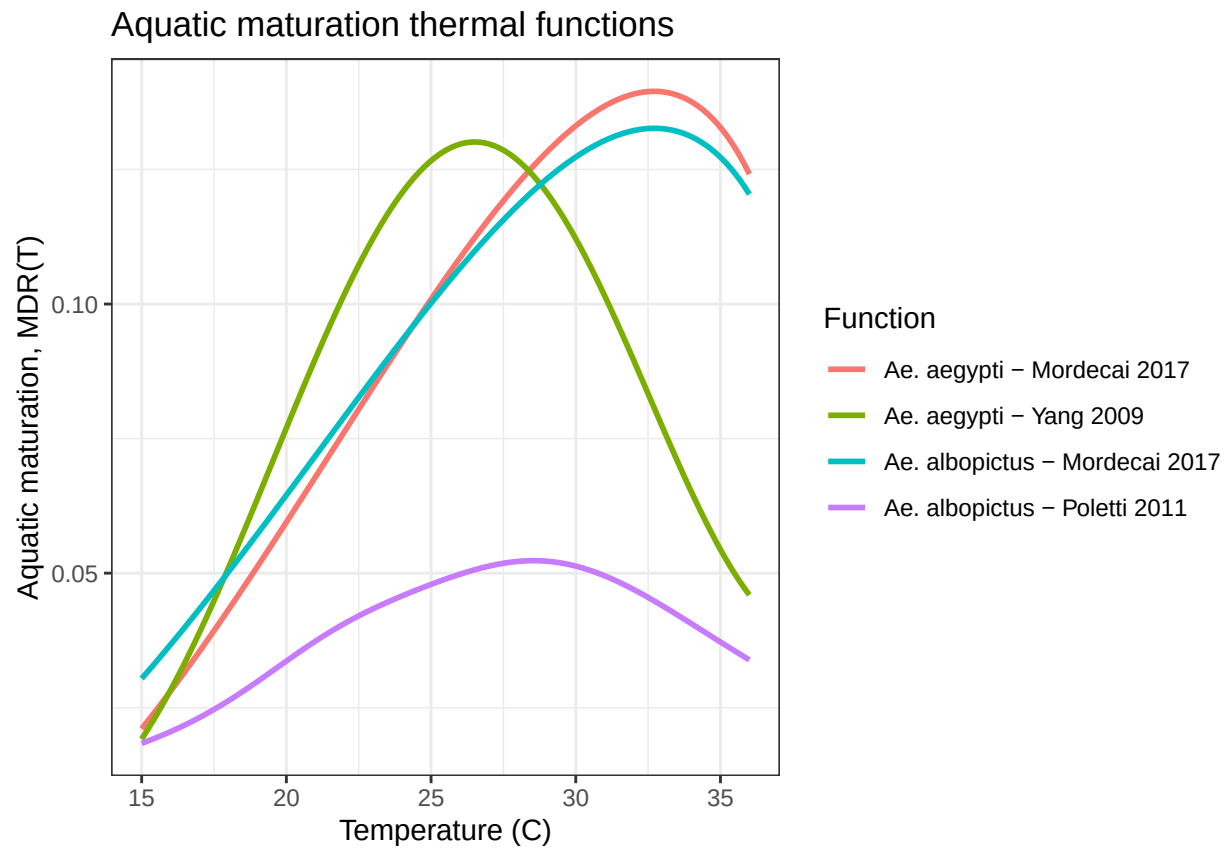


Figure 8: Aquatic maturation thermal functions

Mordecai R0

This section combines the default Mordecai et al. 2017 thermal functions into the baseline temperature-dependent $R_0(T)$ expression.

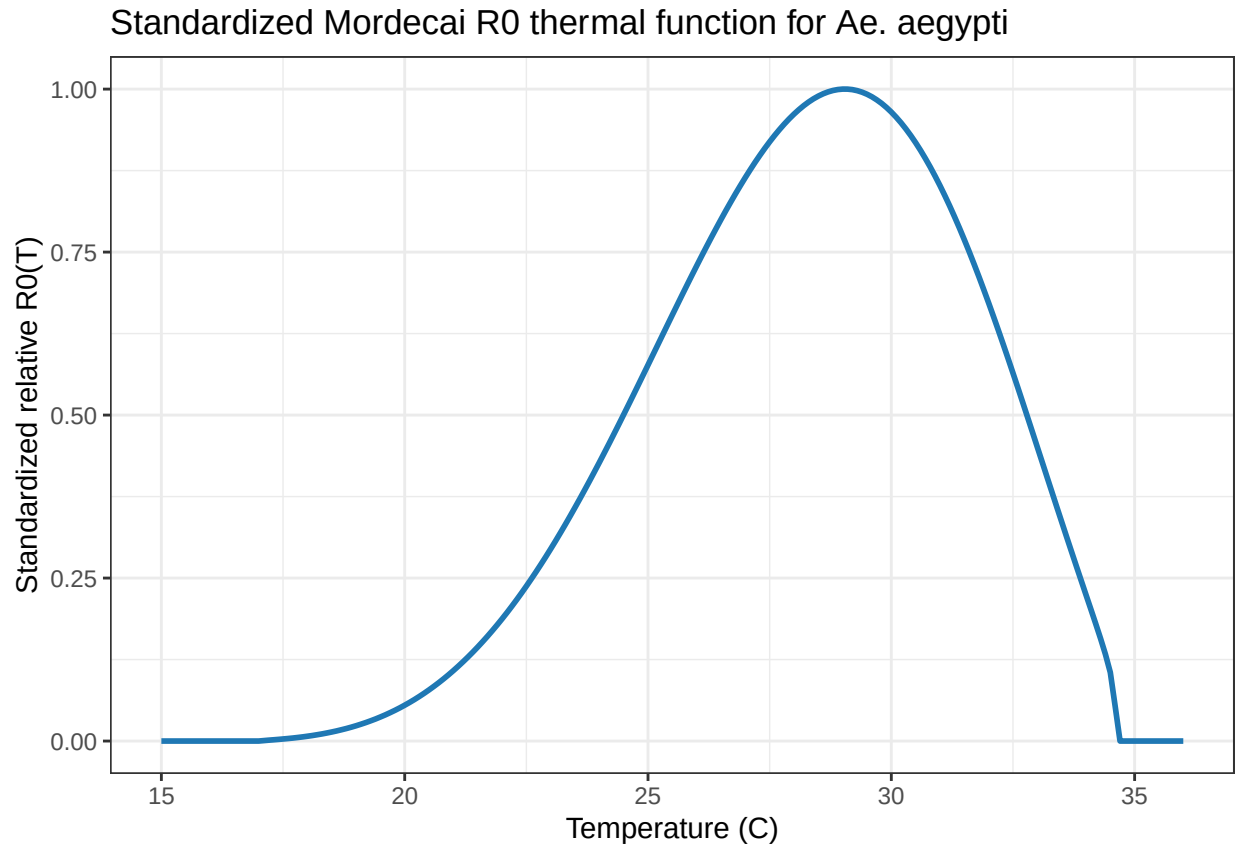


Figure 9: Mordecai R0 thermal functions

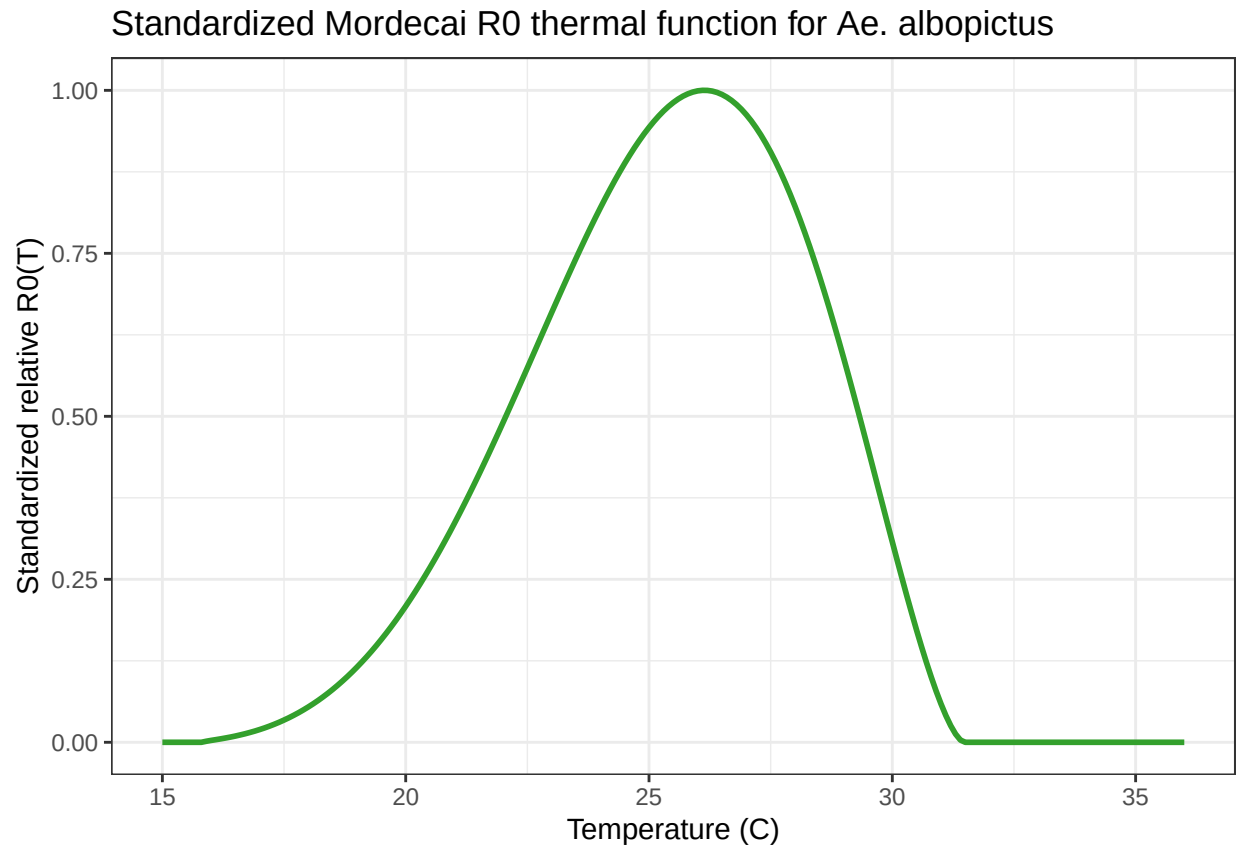


Figure 10: Mordecai R0 thermal functions